

CS 3006

Parallel and Distributed Computing

Lecture 2

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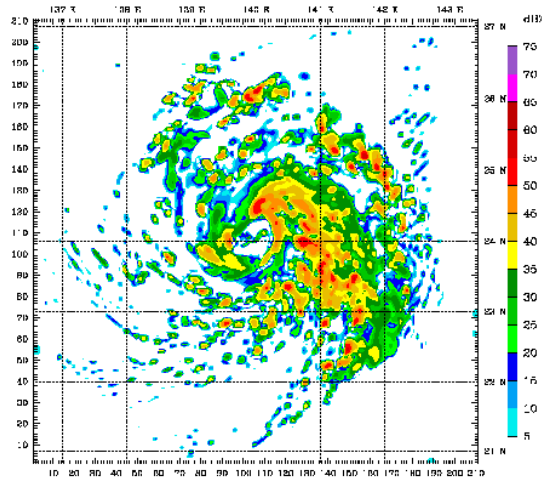
Introduction to Parallel and Distributed Computing

Lecture's Agenda

- **Motivating Parallelism**
- **Systems vs Computing**
- **Parallel vs Distributed Computing**
- **Practical Applications of Parallel and Distributed Computing**
- **Summary**
- **Additional Resources**

Motivation for Parallel and Distributed Computing

- Uniprocessor are **fast** but
 - Some problems require too much computation
 - Some problems use too much data
 - Some problems have too many parameters to explore
- For example
 - Weather simulations, online gaming, web servers, code breaking



Motivating Parallelism

- Developing **parallel hardware and software** has traditionally been time and effort intensive
- If one is to view this in the context of rapidly improving **uniprocessor speeds**, one is tempted to question the need for parallel computing
- Latest trends in hardware design indicate that uniprocessors are not able to sustain the rate of realizable **performance increments** in the future
- A number of fundamental physical and computational **limitations**
- The **emergence** of standardized parallel programming environments, libraries, and hardware have significantly reduced time to develop (parallel) solution

Motivating Parallelism (Cont.)

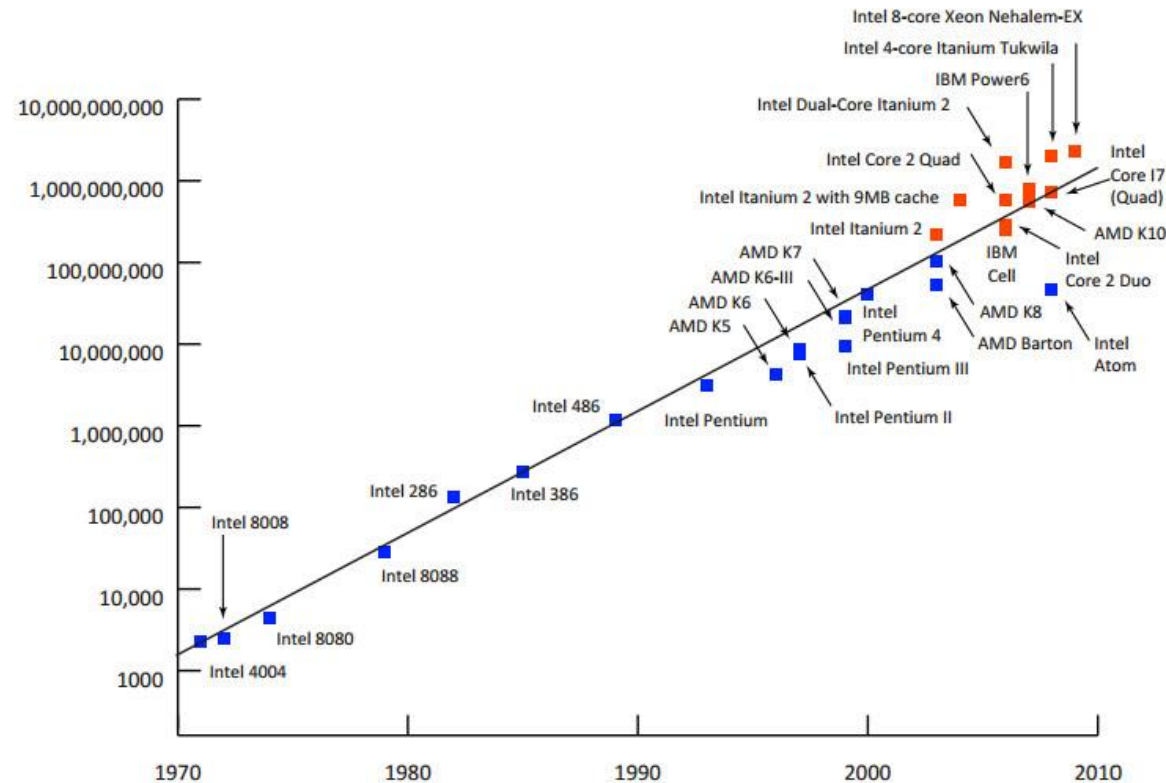
What is Moore's Law?

- Proposed by **Gorden E. Moore** in 1965 and revised in 1975
- It states that [Simplified Version]
 - “Processing speeds, or overall **processing power** for computers will double every 18 months.”
- A more technically correct interpretation
 - “The **number of transistors** on an affordable CPU would double every 18 months.”

Motivating Parallelism (Cont.)

What is Moore's Law? (Cont.)

- Number of transistors incorporated in a chip will approximately **double** every 18 months.



Motivating Parallelism (Cont.)

What is Moore's Law? (Cont.)

- More computational power implicitly **means** more transistors
- Then **why need** second interpretation?
- Let's have a look on **empirical data** from 1970 to 2009
 - In 1970's (i.e., from 1970 to 1979), processor speeds ranged from 740 KHz to 8 Mhz. Difference shows that **both the interpretations** are correct
 - From 2000 to 2009, Speeds ranged from 1.3 GHz to 2.8 GHz
 - ✓ Speed difference is too low but, number of integrated transistors ranged from 37.5 million to 904 million
 - ✓ So, **second interpretation** is more accurate.

Motivating Parallelism (Cont.)

Why doubling the transistors does not doubles the speed?

- The answer is increase in number of transistor per processor is due to **multi-core CPU's**
- In multi-core CPUs, speed will depend on the **proper utilization** of parallel resources
- To follow Moore's law, companies had to:
 - Introduce Ultra Large Scale Integrations (ULSI)
 - ✓ Ultra large-scale integration (ULSI) is the process of embedding millions of transistors on a single silicon semiconductor microchip
 - And multi-core processing era

Motivating Parallelism (Cont.)

Will Moore's law hold forever?

- Adding multiple cores on single chip causes **heat issues**
- Increasing the number of cores, may not be able to increase speeds [Due to **inter-process interactions**]
- Transistors would eventually reach the **limits of miniaturization** at atomic levels

Motivating Parallelism (Cont.)

- So, we must look for **efficient parallel software solutions** to fulfill our future computational needs
- As stated earlier, number of cores on a single chip also have physical and computational **limitations**
- **Solution:**
 - Need to find more **scalable distributed and hybrid solutions**

Motivating Parallelism (Cont.)

The Memory / Disk Speed Argument

- While **clock rates** of high-end processors have increased at roughly 40% per year over the past decade, DRAM access times have only improved at the rate of roughly 10% per year over this interval
- This **mismatch in speeds** causes significant performance bottlenecks
- Parallel platforms provide **increased bandwidth** to the memory system

Motivating Parallelism (Cont.)

The Memory / Disk Speed Argument (Cont.)

- Parallel platforms also provide **higher aggregate caches**
- Some of the fastest growing applications of parallel computing utilize not their raw computational speed, rather their **ability to pump data** to memory and disk faster
 - CPU is fast but memory isn't, which is a **mismatch** in context of parallel programming
 - Solution: Parallel platform (increased **bandwidth** to memory) and combination or collection of caches

Motivating Parallelism (Cont.)

The Data Communication Argument

- As the network evolves, the vision of the Internet as **one large computing platform** has emerged
- In many applications like databases and data mining problems, the **volume of data** is such that they cannot be moved
- Any **analyses** on this data must be performed over the network using parallel techniques

Systems vs Computing

Distributed Systems:

- “A collection of autonomous computers, connected through a network and **distribution middleware**.”
- Enables computers to **coordinate** their activities and to share the resources of the system
- The system is usually perceived as a single, **integrated** computing facility
- Mostly concerned with the **hardware-based** accelerations

Systems vs Computing (Cont.)

Distributed Computing:

- “A specific use of distributed systems to split a large and complex processing into subparts and execute them in **parallel** to increase the productivity.”
- Computing mainly concerned with **software-based** accelerations (i.e., designing and implementing algorithms)

Parallel vs Distributed Computing

Parallel Computing:

- The term is usually used for developing **concurrent solutions** for following two types of the systems:
 - Multi-core Architecture
 - Many core architectures (i.e. CPU's & GPU's)

Distributed Computing:

- The type of computing mainly concerned with **developing algorithms** for the distributed cluster systems
- Here distributed means a **geographical distance** between the computers without any shared-Memory

Parallel vs Distributed Computing (Cont.)

Parallel Computing:

- Shared memory
- Multiprocessors
- Processors share logical address spaces
- Processors share physical memory
- Referred to the study of parallel algorithms

Distributed Computing:

- Distributed memory
- Clusters of workstations
- Processors do not share address space
- There is no sharing of physical memory
- Study of theoretical distributed algorithms or neural networks

Practical Applications of P&D Computing

Scientific Applications:

- Functional and structural characterization of **genes** and proteins
- Applications in **astrophysics** have explored the evolution of galaxies, thermonuclear processes, and the analysis of extremely large datasets from telescope
- Advances in **computational physics and chemistry** have explored new materials, understanding of chemical pathways, and more efficient processes

Practical Applications of P&D Computing (Cont.)

Scientific Applications (Cont.):

- Weather modeling for **simulating** the track of natural hazards like extreme cyclones (storms), flood prediction
- Large Hydron Collider (LHC) at European Organization for Nuclear Research (CERN) **generates petabytes of data** for a single collision

Practical Applications of P&D Computing (Cont.)

Commercial Applications:

- Some of the largest parallel computers power **Wall Street!**
- Data mining analysis for **optimizing business** and marketing decisions
- Large scale servers (**mail and web servers**) are often implemented using parallel and distributed platforms
- Applications such as **information retrieval and search** are typically powered by large clusters

Practical Applications of P&D Computing (Cont.)

Computer Systems Applications:

- Network intrusion detection: a large amount of data needs to be **analyzed** and processed
- Cryptography (the art of writing or solving codes) employs **parallel infrastructures** and algorithms to solve complex codes – SHA 256
- Graphics processing – Each **pixel** represents a binary
- Embedded systems increasingly **rely** on distributed control algorithms - Modern automobiles use embedded systems

Limitations of Parallel Computing

- Parallel computing requires designing the **proper communication and synchronization mechanisms** between the processes and the sub-tasks.
- Exploring the **proper parallelism** from a problem is a hectic process.
- The program must have **low coupling and high cohesion** however it is difficult to create such programs.
- Parallel computing needs relatively **more technical skills** to code a parallel program.

Summary

- **Motivation for Parallel and Distributed Computing**
 - Apps like weather simulations, online gaming, web servers etc. requires PDC
- **Motivating Parallelism**
 - Uniprocessors cannot provide required performance due to their computational limitations
 - **Moore's Law - Number of transistors double every 18 months**
 - ✓ Doubling the transistors does doubles the speed?
 - True for single core chip but not for multi-core CPUs
 - ✓ Will Moore's law hold forever?
 - Heat issues, synchronization issues, limits of miniaturization of transistors

Summary (Cont.)

- **Motivating Parallelism**

- **The Memory / Disk Speed Argument**

- ✓ Speed mismatch between processor and DRAM can be reduced using parallel platforms and aggregated caches

- **The Data Communication Argument**

- ✓ Analyses on large scale data can be performed over the network using parallel techniques

Summary (Cont.)

- **Distributed System vs Distributed Computing**

- **Distributed System** - Collection of autonomous computers, connected through a network and distribution middleware
- **Distributed Computing** - Specific use of distributed systems to split a large problem into subparts and execute them in parallel

- **Parallel Computing vs Distributed Computing**

- **Parallel Computing** - Developing concurrent solutions
- **Distributed Computing** - Autonomous computers on different locations connected via network

Summary (Cont.)

- **Practical Applications of P&D Computing**

- **Scientific Applications** - Characterization of genes and proteins, astrophysics applications, computational physics and chemistry, weather modeling, nuclear physics
- **Commercial Applications** - Processing of stock exchanges, banking sector, business analytics, web servers, mail servers, search engines
- **Computer Systems Applications** - Network intrusion detection, cryptography, graphic processing, embedded systems control

- **Limitations of Parallel Computing**

- **Parallel computing** requires designing proper communication and synchronization mechanisms
- **Parallel program** must have low coupling and high cohesion

Additional Resources

- **Introduction to Parallel Computing** by Ananth Grama and Anshul Gupta

- **Chapter 1: Introduction to Parallel Computing**

- ✓ **1.1 Motivating Parallelism**

- **1.1.1 The Computational Power Argument – from Transistors to FLOPS**
 - **1.1.2 The Memory / Disk Speed Argument**
 - **1.1.3 The Data Communication Argument**

- ✓ **1.2 Scope of Parallel Computing**

Questions?